Setting Up Partitions and Persistent Mounting on AWS EC2.

Objective:

Configure partitions on an AWS EC2 instance and ensure persistent mounting using /etc/fstab.

```
root@ip-172-31-36-123:/home/ubuntu# lsblk
NAME         MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0          7:0    0 25.2M  1 loop /snap/amazon-ssm-agent/7983
loop1          7:1    0 55.7M  1 loop /snap/core18/2812
loop2          7:2    0 55.7M  1 loop /snap/core18/2823
loop3          7:3    0 63.9M  1 loop /snap/core20/2264
loop4          7:4    0 63.9M  1 loop /snap/core20/2318
loop5          7:5    0   87M  1 loop /snap/lxd/27948
loop6          7:6    0   87M  1 loop /snap/lxd/28373
loop7          7:7    0 38.7M  1 loop /snap/snapd/21465
loop8          7:8    0 38.8M  1 loop /snap/snapd/21759
loop9          7:9    0 25.2M  1 loop /snap/amazon-ssm-agent/7993
nvme0n1      259:0    0    8G  0 disk
├─nvme0n1p1  259:1    0  7.9G  0 part /
├─nvme0n1p14 259:2    0    4M  0 part
└─nvme0n1p15 259:3    0  106M  0 part /boot/efi
nvme1n1      259:4    0    8G  0 disk
root@ip-172-31-36-123:/home/ubuntu# exit
exit
ubuntu@ip-172-31-36-123:~$ exit
```

Nvme1n1 is the newly attached volume . and we are going to make partitions on that.

Partition Creation:

Created multiple partitions on the instance using fdisk.
Assigned appropriate sizes to each partition (e.g., 2GB each).

```
Command (m for help): n
Partition type
   p   primary (1 primary, 0 extended, 3 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (2-4, default 2): 2
First sector (4196352-16777215, default 4196352):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (4196352-16777215, default 16777215): +2G

Created a new partition 2 of type 'Linux' and of size 2 GiB.

Command (m for help): n
Partition type
   p   primary (2 primary, 0 extended, 2 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (3,4, default 3): 3
First sector (8390656-16777215, default 8390656):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (8390656-16777215, default 16777215): +2G

Created a new partition 3 of type 'Linux' and of size 2 GiB.
```

```
   e   extended (container for logical partitions)
Select (default p): p
Partition number (3,4, default 3): 3
First sector (8390656-16777215, default 8390656):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (8390656-16777215, default 16777215): +2G

Created a new partition 3 of type 'Linux' and of size 2 GiB.

Command (m for help): n
Partition type
   p   primary (3 primary, 0 extended, 1 free)
   e   extended (container for logical partitions)
Select (default e):

Using default response e.
Selected partition 4
First sector (12584960-16777215, default 12584960):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (12584960-16777215, default 16777215):
```

Filesystem Formatting:

Formatted the partitions using mkfs.ext4 for ext4 filesystems.

> The mkfs command makes a new file system on a specified device. The mkfs command initializes the volume label, file system label, and startup block.

```
root@ip-172-31-36-123:/home/ubuntu# sudo mkfs.ext4 /dev/nvme1n1p1
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 524288 4k blocks and 131072 inodes
Filesystem UUID: fbd4e7f1-f26c-40a0-bb11-0d150a21805e
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

root@ip-172-31-36-123:/home/ubuntu# sudo mkfs.ext4 /dev/nvme1n1p2
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 524288 4k blocks and 131072 inodes
Filesystem UUID: 82055b32-a38b-4066-a41b-8e386e598b38
Superblock backups stored on blocks:
```

Updating /etc/fstab:

Edited the /etc/fstab file using sudo vi /etc/fstab.

Added entries for each partition using UUIDs to ensure persistent mounting.

Example entry: UUID=82055b32-a38b-4066-a41b-8e386e598b38 /mnt/data2 ext4 defaults,nofail 0 2.

```
root@ip-172-31-36-123:/home/ubuntu# sudo mkdir /mnt/data1
root@ip-172-31-36-123:/home/ubuntu# sudo mkdir /mnt/data2
root@ip-172-31-36-123:/home/ubuntu# sudo mkdir /mnt/data3
root@ip-172-31-36-123:/home/ubuntu# sudo mount /dev/nvme1n1p1 /mnt/data1
root@ip-172-31-36-123:/home/ubuntu# sudo mount /dev/nvme1n1p2 /mnt/data2
root@ip-172-31-36-123:/home/ubuntu#
^C
root@ip-172-31-36-123:/home/ubuntu# sudo mount /dev/nvme1n1p3 /mnt/data3
root@ip-172-31-36-123:/home/ubuntu# df -h
Filesystem       Size  Used Avail Use% Mounted on
```

Using sudo blkid command i got each partitions UUID's.

```
/dev/nvme0n1p1: LABEL="cloudimg-rootfs" UUID="db5e27f6-7377-40b0-9756-df259213cbb0" BLOCK_SIZE="4096" TYPE="ext4"
/dev/nvme0n1p15: LABEL_FATBOOT="UEFI" LABEL="UEFI" UUID="692D-8804" BLOCK_SIZE="512" TYPE="vfat" PARTUUID="e556a7
/dev/loop1: TYPE="squashfs"
/dev/nvme0n1p14: PARTUUID="d5a642d2-c9d7-4be1-a293-bcc064177b4f"
/dev/loop8: TYPE="squashfs"
/dev/loop6: TYPE="squashfs"
/dev/loop4: TYPE="squashfs"
/dev/loop2: TYPE="squashfs"
/dev/loop0: TYPE="squashfs"
/dev/loop9: TYPE="squashfs"
/dev/loop7: TYPE="squashfs"
/dev/nvme1n1p2: UUID="82055b32-a38b-4066-a41b-8e386e598b38" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="3530a934-02"
/dev/nvme1n1p3: UUID="4cce72ce-6750-46a2-89b6-8f6ab12fdc17" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="3530a934-03"
/dev/nvme1n1p1: UUID="fbd4e7f1-f26c-40a0-bb11-0d150a21805e" BLOCK_SIZE="4096" TYPE="ext4" PARTUUID="3530a934-01"
/dev/loop5: TYPE="squashfs"
```

As you can see i have added UUID in /etc/fstab

```
tmpfs           383M  4.0K  383M   1% /run/user/1000
root@ip-172-31-36-123:/home/ubuntu# sudo cat /etc/fstab
LABEL=cloudimg-rootfs   /        ext4   discard,errors=remount-ro       0 1
LABEL=UEFI      /boot/efi       vfat    umask=0077      0 1

UUID=82055b32-a38b-4066-a41b-8e386e598b38 /mnt/data2 ext4 defaults,nofail 0 2
UUID=4cce72ce-6750-46a2-89b6-8f6ab12fdc17 /mnt/data3 ext4 defaults,nofail 0 2
UUID=fbd4e7f1-f26c-40a0-bb11-0d150a21805e /mnt/data1 ext4 defaults,nofail 0 2
root@ip-172-31-36-123:/home/ubuntu#
```

After all i did sudo mount -a for checking any errors and rebooted my instance.

Mounting Verification:

Used sudo mount -a to test the /etc/fstab entries for errors.
Verified successful mounting using df -h to check mounted filesystems.

```
Last login: Fri May 31 06:09:26 2024 from 27.6.32.62
ubuntu@ip-172-31-36-123:~$ sudo su
root@ip-172-31-36-123:/home/ubuntu# df -h
Filesystem       Size  Used Avail Use% Mounted on
/dev/root        7.6G  3.6G  4.0G  48% /
tmpfs            1.9G     0  1.9G   0% /dev/shm
tmpfs            766M  896K  765M   1% /run
tmpfs            5.0M     0  5.0M   0% /run/lock
efivarfs         128K  4.1K  119K   4% /sys/firmware/efi/efivars
/dev/nvme1n1p3   2.0G   24K  1.8G   1% /mnt/data3
/dev/nvme1n1p2   2.0G   24K  1.8G   1% /mnt/data2
/dev/nvme0n1p15  105M  6.1M   99M   6% /boot/efi
/dev/nvme1n1p1   2.0G   24K  1.8G   1% /mnt/data1
tmpfs            383M  4.0K  383M   1% /run/user/1000
root@ip-172-31-36-123:/home/ubuntu# sudo cat /etc/fstab
LABEL=cloudimg-rootfs   /        ext4   discard,errors=remount-ro
LABEL=UEFI      /boot/efi       vfat    umask=0077      0 1

UUID=82055b32-a38b-4066-a41b-8e386e598b38 /mnt/data2 ext4 defaults,
```

System Reboot Testing:

Rebooted the EC2 instance to verify that the mounts persist after a reboot.
Checked mounts again post-reboot to ensure persistence.

```
0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Fri May 31 06:09:26 2024 from 27.6.32.62
ubuntu@ip-172-31-36-123:~$ sudo su
root@ip-172-31-36-123:/home/ubuntu# df -h
Filesystem       Size  Used Avail Use% Mounted on
/dev/root        7.6G  3.6G  4.0G  48% /
tmpfs            1.9G     0  1.9G   0% /dev/shm
tmpfs            766M  896K  765M   1% /run
tmpfs            5.0M     0  5.0M   0% /run/lock
efivarfs         128K  4.1K  119K   4% /sys/firmware/efi/efivars
/dev/nvme1n1p3   2.0G   24K  1.8G   1% /mnt/data3
/dev/nvme1n1p2   2.0G   24K  1.8G   1% /mnt/data2
/dev/nvme0n1p15  105M  6.1M   99M   6% /boot/efi
/dev/nvme1n1p1   2.0G   24K  1.8G   1% /mnt/data1
tmpfs            383M  4.0K  383M   1% /run/user/1000
root@ip-172-31-36-123:/home/ubuntu# sudo cat /etc/fstab
LABEL=cloudimg-rootfs   /        ext4   discard,errors=remount-ro
LABEL=UEFI      /boot/efi       vfat    umask=0077      0 1
```

Final Verification:

Confirmed that all partitions are correctly mounted and usable post-reboot.
Ensured no errors or issues in the mount configurations.